

Here is the finalized Software Requirements Specification (SRS). I have explicitly defined that managers are core members with elevated privileges, integrated their member functionalities into their dashboards, and added the **Manager Election & Rotation System** while keeping all your previous requirements intact.

1. User Roles & Authorization Matrix

The system uses Role-Based Access Control (RBAC).

Important Note: All Admins and Managers are fundamentally members of the mess. They maintain their own personal balances, toggle their own meals, and pay fixed costs exactly like General Members. Their roles simply grant them elevated administrative privileges.

Feature / Action	Super Admin	Finance Manager	Meal Manager	General Member
Delete Mess / Remove Members	Yes	No	No	No
Finalize Manager Roles	Yes	No	No	No
Input Daily Bazaar/Fixed Bills	Yes	Yes	No	No
Approve Member Deposits	Yes	Yes	No	No
Close the Month	Yes	Yes	No	No
Set Meal Lock Times	Yes	No	Yes	No
Override/Edit Others' Meals	Yes	No	Yes	No

Create Polls (Meals & Elections)	Yes	No	Yes	No
Post Notices/Announcements	Yes	Yes	Yes	No
Update Own Meals & Vote	Yes	Yes	Yes	Yes
Mark Inventory as "Low Stock"	Yes	Yes	Yes	Yes

2. Role-Specific Dashboards

The UI adapts based on the user's role. Every manager dashboard inherently includes the General Member features so they can manage their own daily needs.

Super Admin Dashboard

- **Top Card:** Mess Overview (Total Members, Active Managers).
- **Quick Actions:** Manage Members, Trigger Manager Elections, Mess Settings.
- **Personal View:** Includes personal meal status and financial balance.

Finance Manager Dashboard

- **Top Card:** Pending Approvals (e.g., Unverified Member Deposits).
- **Quick Actions:** Add Daily Bazaar, Add Fixed Bill, Close Month.
- **Overview:** Total Mess Balance vs. Total Expenses this month.
- **Personal View:** Includes personal meal status and financial balance.

Meal Manager Dashboard

- **Top Card:** Today's Meal Summary (Total Breakfast, Lunch, Dinner counts for the mess).
- **Quick Actions:** Lock Meals, Override Meals, Create Special Meal Poll, Set Today's Menu.
- **Personal View:** Includes personal meal status and financial balance.

General Member Dashboard

- **Top Card:** Personal Financial Standing (Current Balance & Live Meal Rate).
- **Quick Actions:** Toggle Today's Meals (Breakfast, Lunch, Dinner).
- **Notice Section:** Pinned announcements from managers.

- **Red Zone Indicator:** The top card turns red if the balance falls below the minimum threshold.

3. Manager Election & Rotation System

To handle the shifting responsibilities in a mess, the app includes a democratic rotation system.

- **End-of-Month Election:** A few days before the month ends, the app automatically generates an election poll (or the Super Admin triggers it).
- **Voting Mechanism:** All members vote for the upcoming month's Finance Manager and Meal Manager from a list of eligible members.
- **Role Handover:** Upon triggering the "Close Month" function, the system automatically assigns the Manager roles to the election winners for the new month, revoking the privileges from the previous managers.

4. Authentication & Onboarding

- **Registration/Login Flow:** Users register and log in using an Email/Password or Phone Number/Password combination.
- **Profile Setup:** Name, phone number, emergency contact, and an optional profile picture.
- **Mess Creation & Joining:**
 - Super Admin registers, creates a "Mess Profile," and generates a unique Invite Code.
 - New users register, select "Join a Mess," and input the Invite Code.
 - Super Admin must approve the join request before the user is admitted.

5. Meal Management Module

- **Daily Meal Toggling:** Members can enable or disable their standard meals.
- **Fractional & Guest Meals:** Support for 0.5 (half) meals and Guest Meals. Guest meals increase the total mess meal count, but the financial cost is billed entirely to the host member.
- **Automated Meal Locking:** Meal Manager sets cut-off times. The app disables meal toggling for that specific meal after the deadline.
- **Manager Override:** The Meal Manager can manually update a member's meal count after the lock time if necessary.
- **Vacation Mode:** Members can select a date range when traveling home. The app automatically disables all meals for those dates.
- **Ramadan Mode:** A system-wide toggle activated during the month of Ramadan.
 - Automatically shifts meal categories from "Breakfast, Lunch, Dinner" to "**Sehri, Iftar, Dinner.**"
 - Allows custom lock times specific to Ramadan (e.g., locking Sehri the night before).

6. Advanced Meal Features (Feasts)

- **Special Meal Polls:** Meal Manager can create polls for events (e.g., Friday Biryani, Eid Feast).
- **Integrated Special Meal Cost:**
 - The expense for a special meal is recorded with regular bazaar expenses but tagged as a "Special Event."
 - The cost is dynamically divided *only* among the members who opt in to eat (including those who didn't vote but later joined in).
 - This cost is seamlessly added to those specific members' individual month-end calculations, bypassing the regular daily meal rate.
- **Menu Display:** A text section controlled by the Meal Manager to display today's cooking menu.

7. Financial Management & Bazaar

- **Dynamic Meal Rate Calculation:** The app calculates and displays the live meal rate in real-time.
 - *Formula: Total Bazaar Cost / Total Standard Meals Consumed.*
- **Daily Bazaar Entry:** Finance Manager logs daily shopping (Date, Amount, optional photo of the receipt).
- **Fixed Bills Management:** Input for monthly shared costs: Rent, Maid (Bua), WiFi, Electricity, Gas, Water, and Garbage.
- **Deposit Logging:** Members hand over cash/mobile banking funds. The Finance Manager logs it in the system, reflecting immediately in the member's account.

8. Due Tracking & Month-End Auditing

- **Real-Time Balance Tracker:** Members can always see their current estimated standing.
 1. *Calculation: Total Deposits - [(Total Personal Meals × Live Meal Rate) + (Fixed Bills / Member Count) + Personal Special Meal Costs].*
- **Due Tracker (The "Red Zone"):** If a member's balance drops below a predefined threshold (e.g., 500 BDT), their dashboard turns red, signaling a deposit is due.
- **Month Closing:** The Finance Manager or Super Admin triggers the "Close Month" function:
 1. Archives the month's data into a summary report.
 2. Resets all meal counts and bazaar logs to zero.
 3. Carries over the final balances (positive or negative) as the Opening Balance for the new month.
 4. *Executes the role handover to the newly elected managers.*
- **Audit Trail:** A read-only "Recent Activity" feed showing financial and administrative changes (e.g., deposits logged, meal rates updated) for transparency.

9. Mess Operations & Lifestyle

- **Digital Notice Board:** Pinned announcements at the top of the dashboard. Only Managers and Admins can post; Members can read.
- **Bazaar Roster:** A digital calendar showing whose turn it is to do the daily grocery shopping.
- **Inventory Tracker:** A simple checklist for shared bulk items (Rice, Oil, Onion, Gas). *Any user* can flag an item as "Low Stock," alerting the next assigned shopper.

10. Notification Engine

- **Time-Sensitive Reminders:** "Lunch input locks in 30 minutes!"
- **Financial Alerts:** "Your 2000 BDT deposit was approved," or "Low Balance Alert: Please deposit money."
- **Action Required:** "New Poll created - Vote Now!", "Manager Elections are live!"
- **System Events:** "The month has been closed. Tap to view your final invoice."

11. Non-Functional Requirements (NFRs)

- **Bilingual UI:** Toggle between English and **Bangla**.
- **Offline Caching:** Users can open the app, view their current balance, meal status, and notices even without an active internet connection.
- **Concurrency & Performance:** The dynamic meal rate and user balances must recalculate instantly without UI lag when new data is entered.
- **Platform constraints:** Built in Android Studio, optimized for modern Android architectures.